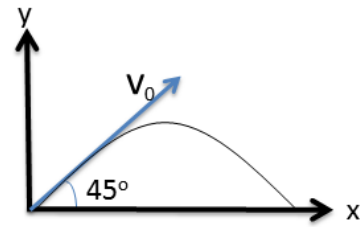
PHY101: Introduction to Physics-1

Monsoon, 2017

Tutorial-13

1. A particle is located at the vector position **r** = (***i*** + 3***j***) m, and the force acting on it is **F** = (3***i*** - 2***j***) N. What is the torque about the point having coordinates (0, 6) m?

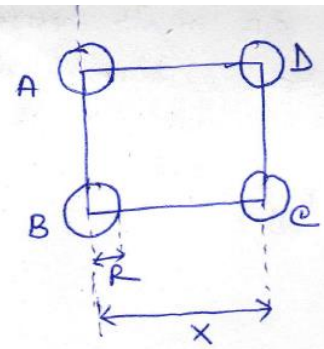
2. A particle of mass *m* is projected at time *t* = 0 from a point *P* with a speed v_0 at an angle of 45° to the horizontal. Find the magnitude and the direction of the angular momentum of the particle about the point *P* at time $t = v_0/g$.



3. Find the moment of inertia of a uniform sphere of mass *M* and radius *R* around an axis through the center.

4. Four spheres each of radius *R* and mass *M* are placed with their centres on the four corners of a square of side *X*. Calculate the moment of inertia of the system about one side of the square.

(i) Let us calculate the M.I. of the system about the side AB of the square ABCD.



M.I. of a sphere about an axis along its diameter $= \frac{2}{5}MR^2$.

So, M.I. of the sphere D about AB axis, $I_D = \frac{2}{5}MR^2 + Mx^2$ (Parallel axes theorem)

Similarly, M.I. of the sphere C about AB axis,

$$I_C = \frac{2}{5}MR^2 + Mx^2$$

M.I. of sphere A about AB axis, $I_A = \frac{2}{5}MR^2$

B " " " $I_B = \frac{2}{5}MR^2$

$\therefore$ M.I. of the system $= 2.\left(\frac{2}{5}MR^2 + Mx^2\right) + 2\left(\frac{2}{5}MR^2\right)$

$$= \frac{4}{5}MR^2 + 2Mx^2 + \frac{4}{5}MR^2$$

$$= \frac{8}{5}MR^2 + 2Mx^2$$